

CAREER: Building Physics-Inspired 3D Scene Representations For Efficient And Interpretable Machine Perception

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Overview

Modern AI models for 3D scene understanding are data-hungry, computationally expensive, and uninterpretable because they ignore the physics governing the phenomena they model. **This proposal’s central thesis: physics, embedded as inductive biases, produces smaller, faster, and more interpretable AI systems.** We replace the nonlinear Eikonal PDE underlying signed distance function (SDF)-based reconstruction with Poisson’s equation—a linear surrogate with closed-form solutions—and parameterize those solutions with compact deep networks. The model learns parameters of an analytically derived solution rather than regressing the entire solution space; it requires less data, fewer parameters, and no competing loss terms. This is not a minor implementation improvement—it is a **paradigm shift**: from fitting black-box functions to physics space, to parameterizing known physics within machine learning.

Aim 1 (Years 1–2): Establish foundations of electrostatic 3D representations—differentiable charge interactions, stable training objectives, compression-fidelity tradeoffs, and uncertainty-aware fields—benchmarked against NeRF and Gaussian Splatting with full open-source release. **Aim 2 (Years 2–4):** Extend to time-varying scenes and embodied perception. Time-evolving charges define a *dynamic field memory*, enabling real-time 3D reasoning for UAV navigation, robotic manipulation, event cameras, and dynamic obstacle avoidance. **Aim 3 (Years 3–5):** Deploy in high-impact domains: wildfire plume reconstruction, infrastructure inspection, environmental monitoring, and medical volumetric imaging—all scenarios characterized by sparse, noisy, or asynchronous sensor data where physics-constrained representations have a decisive advantage over pure data-driven methods.

Intellectual Merit

The proposal makes three fundamental contributions. **(1) A new class of physically grounded, efficient, and interpretable 3D scene representations.** Linearizing the governing PDE unlocks analytical solution techniques—Green’s functions, Fourier and Laplace transforms, separation of variables—categorically unavailable for nonlinear formulations. The resulting representations are provably smooth, globally unique, and *decomposable into interpretable geometric primitives via superposition*—a structural property with no counterpart in NeRF or Gaussian Splatting. **(2) A scalable theory of dynamic field memory.** Time-varying charge distributions yield a principled, physics-consistent mechanism for representing motion without ad hoc temporal smoothing or auxiliary memory modules. **(3) Unified perception for active and embodied agents.** Connecting field representations to UAV-based atmospheric sensing, event cameras, and robotic planning produces a single coherent perception framework grounded in field theory rather than heuristic engineering. Preliminary results confirm that Poisson-based representations match SDF reconstruction accuracy with significantly fewer parameters and without Eikonal regularization.

Broader Impacts

Curriculum. The PI will develop *Physics-Inspired AI for Vision and Robotics*—a modular course sequence covering potential fields in ML, geometric deep learning, 3D representations, and differentiable simulation—deployed at UTA, a Hispanic-Serving Institution serving 40,000+ students.

The PhysicsAI Scholars Program (a structured expansion of the PI’s existing UR2PhD pipeline) will provide research pathways for first-generation, Hispanic, transfer, and women students through

summer 3D vision projects, leveraging established CASHI mentoring infrastructure already in place.

K-12 Outreach. Hands-on workshops—*How AI Sees 3D Worlds*—use electrostatic analogies (fields, charges, forces) to make 3D AI accessible to middle and high school students. An open interactive simulator lets students place Gaussian charges and watch 3D shapes emerge in real time.

Open Science. All datasets, trained models, educational modules, and cloud demos will be publicly released, extending impact well beyond UTA.

Societal Impact. The framework directly targets wildfire disaster response and environmental monitoring—domains where accurate 3D reconstruction under extreme sensor uncertainty is a public safety imperative, not an academic benchmark.